

Coding & Solution

Date	3 July 2026
Team ID	
Project Name	Voice-Based Communication Understanding Analyzer
Maximum Marks	5 Marks

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The **Voice-Based Concept Understanding Analyser (VBCUA)** is implemented using Python and Streamlit. The application integrates speech recognition, semantic similarity analysis, audio feature extraction, intelligent scoring, waveform visualization, and PDF report generation into a modular and user-friendly system. The solution addresses the problem of objectively evaluating spoken conceptual understanding by providing automated assessment and real-time feedback.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/muppagownisaikrishna/Vbcua
Programming Language(s)	Python
Framework(s) Used	Streamlit
Key Features Implemented	• Audio Upload (WAV/MP3) • Speech-to-Text using OpenAI Whisper • Semantic Similarity Analysis using Sentence-BERT • Audio Feature Extraction using Librosa • Waveform Visualization using Matplotlib • Intelligent Scoring Engine • Qualitative Feedback Generation • Downloadable PDF Report using ReportLab
Pending / Incomplete Features	• User Authentication • Database Integration for storing evaluation history • Multi-user support • Cloud Deployment and User Dashboard

Setup / Run Instructions	1. Install Python 3.10+. 2. Install dependencies using <code>pip install -r requirements.txt</code>. 3. Configure the <code>.env</code> file (if API keys are required). 4. Install FFmpeg and add it to the system PATH. 5. Run the application using <code>streamlit run app.py</code>. 6. Open the provided localhost URL in a web browser.
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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The application follows a modular architecture with separate Python modules for speech transcription, semantic analysis, audio feature extraction, waveform generation, scoring, and report generation.
- AI models including OpenAI Whisper and Sentence-BERT are integrated to provide accurate speech recognition and conceptual similarity evaluation.
- The Streamlit-based interface offers an interactive and user-friendly experience for students and educators.
- Waveform visualization and PDF report generation improve result interpretation and documentation.
- The project is designed for future enhancements such as database integration, user authentication, cloud deployment, and performance optimization.